

ICT ACADEMY – CYBER SECURITY PROGRAM

Day 2 – Operating Systems, Linux & Cloud Connectivity

Learning Log / GitHub Documentation

Session Details

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Day	Day 2
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Hands-on Platform	Amazon Web Services (AWS)
Focus Area	Operating Systems, Linux fundamentals and cloud connectivity

Day 2 Focus

Understand what an operating system does and how Linux fits into the OS landscape, learn how systems connect to each other (SSH, RDP), get familiar with Linux shells, basic commands and package managers, and review core cloud computing concepts and terminology across AWS, Azure and Google Cloud.

1. Operating System Overview

An Operating System (OS) is system software that manages computer hardware and provides services and an interface for users and applications.

Major OS Responsibilities

Process Management	Creates, schedules and controls processes.
Services	Provides common system services to applications and users.
Memory Management	Allocates and manages RAM and virtual memory.
Searching / File Management	Helps locate, organize and access files and resources.
Package Management	Downloads, installs, updates and removes software packages.
Security Management	Protects systems using authentication, permissions, antivirus/security controls.
Scripting	Automates administration and repetitive tasks.
Extra Security	Additional hardening, monitoring and access controls.
Server Management	Manages services, users, resources and server operations.
Troubleshooting	Identifies and resolves OS, application, network and hardware problems.

2. OS Components

OS / Kernel	Core software that manages hardware and system resources.
Shell (Command Line)	Command-based interface between the user and OS.
Interpreter	Executes instructions/scripts directly.
Compiler	Translates source code into machine/object code.
Hardware	Physical components such as CPU, memory, storage and I/O devices.

3. Common OS Families / Examples

Family / OS	Examples / Notes
Linux	Red Hat family: RHEL, CentOS, Fedora. Debian family: Debian, Ubuntu, Kali, etc.
Windows	Microsoft desktop/server operating system.
macOS	Apple desktop operating system.
Unix	Unix and Unix-like operating systems.
Android	Mobile OS based on the Linux kernel.

4. Booting Process – Basic Idea

Windows	Firmware/boot manager loads Windows boot components, then the kernel and required services/drivers start.
Linux	Firmware loads a bootloader (commonly GRUB), which loads the Linux kernel and starts the system.
macOS	Firmware/boot process loads macOS components and the kernel, followed by system services.
Unix / Android	Follow platform-specific firmware/bootloader stages before loading the kernel and system services.

5. Linux Overview

Linux is a family of Unix-like operating systems built around the Linux kernel, commonly grouped into the Red Hat family (RHEL, CentOS, Fedora) and the Debian family (Debian, Ubuntu, Kali, etc.).

Linux learning topics: connectivity, basic commands, package management, security, scripting, services, storage, users/permissions and troubleshooting.

6. Connecting to Linux from Windows

A Windows computer can connect to a Linux machine using a third-party tool such as PuTTY, or a command-line client such as PowerShell / Windows Terminal.

Protocol: SSH (Secure Shell) • **Transport:** TCP • **Default port:** 22

The client connects to the Linux server's IP address/hostname and authenticates with a username/password or key.

Windows PC → PuTTY / PowerShell → SSH/TCP → Port 22 → Linux Server

7. Windows Remote Desktop (RDP)

For graphical remote access to Windows, the notes identify RDP (Remote Desktop Protocol).

Protocol: RDP • **Transport:** TCP • **Default port:** 3389

Used for a graphical user interface (GUI) session to a Windows computer/server.

8. Linux ↔ macOS / Other Unix-like Systems

macOS → Linux	SSH over TCP, normally port 22.
Linux → macOS	SSH over TCP, normally port 22, when SSH access is enabled.
Linux → Linux	SSH over TCP, normally port 22.
Linux → Windows	SSH when an SSH server is enabled; RDP may be used for GUI access.

9. What is a Shell in Linux?

A shell is a command-line interface (CLI) that allows a user to communicate with the operating system by typing commands. It can also execute scripts.

Common Linux Shells		
Shell	Full name / type	Common use
sh	Bourne shell	Basic Unix/Linux scripting and compatibility.
bash	Bourne Again Shell	Common Linux shell; administration and scripting.
zsh	Z shell	Interactive terminal and scripting; popular on modern systems.
ksh	Korn shell	Enterprise/Unix scripting.
cs	C shell	C-like syntax; historical Unix systems.
tcsh	TENEX C shell	Enhanced C shell with interactive features.
fish	Friendly Interactive Shell	Beginner-friendly interactive terminal.
dash	Debian Almquist Shell	Lightweight shell and scripting.
ash	Almquist shell	Embedded/minimal Unix-like systems.

10. Basic Linux Commands	
Command	Purpose
pwd	Shows the current working directory.
ls	Lists files and directories.
cd	Changes the current directory.
mkdir	Creates a directory.
touch	Creates an empty file or updates a file timestamp.
cp / mv	Copies / moves files or directories.
rm	Removes files; use carefully.
cat	Displays file contents.
grep	Searches text for matching patterns.
ps / top	Views running processes / system activity.
chmod / chown	Changes permissions / ownership.
ssh	Connects to a remote system securely.

11. Package Management
- A package manager downloads, installs, updates and removes software packages and their dependencies. Debian/Ubuntu: APT is commonly used (e.g. apt update, apt install package). RHEL/Fedora family: DNF is commonly used (e.g. dnf install package). - The handwritten notes specifically emphasize package downloading and installation.

12. Security & Administration
- Security management includes authentication, authorization, permissions, patching, firewalls, antivirus/security tools and monitoring. - Scripting can automate routine administration and troubleshooting. - Server management covers services, users, resources, connectivity and system health.

13. What is Cloud?
Cloud computing is the delivery of computing resources and services over a network, commonly the Internet. The notes describe the cloud as a collection of domains/resources: networking, database, storage, virtualization, containerization, security, IAM, troubleshooting, automation and reverse engineering.

14. Cloud Provider Terms

Provider	Term used in notes	General idea
AWS	Services	Cloud services/resources are consumed through AWS.
Microsoft Azure	Resources	Cloud resources are provisioned and managed through Azure.
Google Cloud	Engines	Cloud compute and other resources are provided through Google Cloud.

15. Quick Connectivity Reference

Connection	Common method	Protocol	Port
Windows → Linux	PuTTY / PowerShell / Windows Terminal	SSH over TCP	22
Windows → Windows	Remote Desktop	RDP over TCP	3389
Linux → Linux	SSH	SSH over TCP	22
macOS → Linux	SSH	SSH over TCP	22
Linux → macOS	SSH	SSH over TCP	22

16. Exam / Interview Revision Points

- **OS:** manages processes, memory, files, devices, security and services.
- **Shell:** CLI used to communicate with the OS and run commands/scripts.
- **SSH:** secure remote shell; normally TCP port 22.
- **RDP:** Windows graphical remote desktop; commonly TCP port 3389.
- **Linux families:** Red Hat family and Debian family are major distribution families.
- **Package management:** use the distribution's package manager to install/update/remove software.
- **Cloud:** on-demand computing resources/services such as networking, storage, databases, virtualization, security and automation.

Architecture & Remote Access Flow

Hardware → Kernel/OS → Shell & System Services → Applications → User

Client → Network → Protocol (SSH/RDP) → Port → Server → Authentication → Session

17. What I Learned on Day 2

1. Understood the core responsibilities of an operating system, from process and memory management to security and troubleshooting.
2. Identified the main OS components: kernel, shell, interpreter, compiler and hardware.
3. Learned the major OS families and how Linux distributions split into Red Hat and Debian families.
4. Understood the basic boot sequence for Windows, Linux, macOS and Unix/Android.
5. Learned how Windows connects to Linux using SSH over TCP port 22.
6. Learned how Windows Remote Desktop (RDP) provides graphical access over TCP port 3389.
7. Reviewed how macOS, Linux and other Unix-like systems connect to each other using SSH.
8. Learned about common Linux shells (bash, zsh, ksh and others) and their typical use cases.
9. Practiced identifying core Linux commands for navigation, file handling and permissions.
10. Understood how package managers (APT, DNF) install and maintain software.
11. Reviewed core cloud computing concepts and how AWS, Azure and Google Cloud each describe their resources.